Global wheat full semantic segmentation using deep learning methods

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***Abstract*—The proper identification of plant structures such as spikes, stems, and leaves is key to current analyses of agriculture and precision farming. This project is a supervised deep learning semantic segmentation of wheat images using a DeepLabV3 model based on a ResNet-50 backbone. The model is trained on RGB images with pixel-wise annotated masks to categorise each pixel as background, spike, stem, or leaf. The pipeline consists of preprocessing, model training, prediction, and evaluation based on the resulting standard segmentation metrics: Pixel Accuracy, Intersection over Union (IoU), and Dice Score. Experimental results show that the model successfully recovers large plant structures, but it is difficult to recover thin parts, such as stems. The suggested solution provides a good foundation for further refinements using more advanced methods, such as class balancing and semi-supervised learning.**

***Index Terms*—Semantic Segmentation, Deep Learning, Wheat Phenotyping, DeepLabV3, ResNet-50, Pixel-wise Classification, Precision Agriculture**

I. Introduction

FUNDAMENTAL requirements for modern precision agriculture are better yield estimation, resource management and disease detection which eventually is an accurate crop monitoring. Among staple crops, wheat plays a critical role in global food security, making its monitoring particularly important. With the rapid advancement of computer vision and deep learning, automated analysis of agricultural images has become a powerful alternative to traditional manual methods.

Sematic Segmentation allows pixel-wise interpretation of the images and is therefore very helpful in agricultural tasks like plant structure analysis and crop yielding. Segmentation enables the accurate identification of components of various plants within a single image, unlike conventional object detection methods.

The work is devoted to dividing the wheat images into four categories: background, spike, stem and leaf, with the help of the deep learning-based methodology. The mapping between input images and segmentation masks is learnt using a pretrained DeepLabV3 model with a ResNet-50 backbone. This deployment is the supervised learning phase of a bigger segmentation pipeline, and a robust starting point for further advancement.

II. Methodology

The supervised deep learning framework proposed to classify wheat images pixel by pixel is based on the proposed methodology. The pipeline comprises multiple steps: data preparation, model design, training, prediction, and evaluation.

A. Data Preparation

The data sample used in this project consists of RGB wheat images and their respective ground-truth segmentation masks. The masks themselves have pixel-level annotations, and each pixel is labelled with a class_id indicating a particular plant part. The data is grouped into two folders: the first for images, and the second for the masks that correspond to them. There is an image titled ”Mask” with the same name.

Preprocessing is performed to ensure the data are consistent and compatible with the neural network before feeding them into the model. All pictures are adjusted to a fixed size of 512 x 512 pixels. This standardisation enables the model to handle images in batches and ensures consistent spatial size across the dataset.

The pixel values of the pictures are normalised to 0-1 by dividing by 255. This normalisation helps stabilise training and enhances convergence in the optimisation process. The photos are then transformed into a channel-first image, as expected by PyTorch.

The nearest-neighbour interpolation is also used to resize the segmentation masks to the same resolution. This step is significant because masks have discrete class labels, and any other interpolation method can introduce false label values. These masks are converted to integer tensors, with each pixel value directly representing a class index.

B. Model Architecture

The model used in this work is DeepLabV3 with ResNet-50. The architecture is popular in semantic segmentation processes because of its capabilities to represent local and global contextual details.

The backbone is ResNet-50, which serves as a feature extractor. It uses the input image to pass through multiple convolutional layers, producing a hierarchical visual feature representation. Simple patterns, such as edges and textures, are captured by higher layers, whereas more complex patterns, such as the shape and arrangement of plants, are captured by lower layers.

These extracted features are then fed into the DeepLabV3 module, which uses multi-scale context aggregation to improve segmentation accuracy. This is very helpful, especially in pictures of agriculture, where objects can vary in size and even overlap.

The last layer of the model is adjusted to produce four channels, corresponding to four segmentation classes. Each

channel is the confidence of a given model to a particular class at a given pixel position.
Pretrained weights are used to initialise the model, enabling it to leverage prior knowledge from large-scale image datasets. This greatly enhances the efficacy and performance of training, particularly when the available dataset is smaller.

C. Training Stratergy

The model is also trained supervised through mini-batch gradient descent. A DataLoader is used to load the dataset, and batches are created and shuffled to ensure random training.
Each time training occurs, a set of images is fed through the model, producing predictions. These predictions are composed of the class scores for each pixel. The ground-truth masks are then compared with the predicted results using a loss function. The training objective is the Cross-Entropy Loss. This loss measure assesses the closeness between the predicted class distributions and the actual class labels at each pixel. When the model has high probability of the correct class, the loss is low, otherwise the loss is high.
Adam optimiser updates model parameters by adjusting weights based on the calculated gradients. Adam is selected as it gives adaptive learning rates and converges more quickly than conventional optimisation techniques.
The repeat is performed several times until the training process is completed. The model is trained on the entire dataset at each epoch. Over time, the model will be able to reduce losses and improve its segmentation predictions.

D. Prediction and Visualization

The model is then used to generate segmentation masks for the input images after training. The model produces class scores for each pixel, and the overall prediction is the class with the highest score.
The results are then transformed into colour-coded images of the predicted masks, with a mapping for each class, to make the results interpretable. This enables to visually compare the foreseen segmentation and ground-truth mask.
The use of visualisation is very important for analysing segmentation performance because it provides insight into how the model captures structural features, boundaries, and thin plant parts.

E. Evaluation metrics

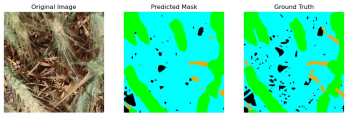
Standards are used to measure the model's performance.
Pixel Accuracy: This measures the ratio of correctly classified pixels across the entire image. Although it is easy to interpret, this might not capture performance when class distributions are imbalanced.
The overlap between the predicted and ground-truth regions for each class is measured using intersection over Union (IoU). It provides a better indication of segmentation quality, particularly for individual classes.
Mean IoU (mIoU) will be computed as an average of IoU values over all the classes. This measure will ensure that performance is measured consistently across categories.
Another useful measure is the Dice Score, which determines the similarity between the predicted and ground-truth areas. It can also be applied to very thin or small structures (such as stems), which are often harder to divide into sections.

F. Methodological Intuition

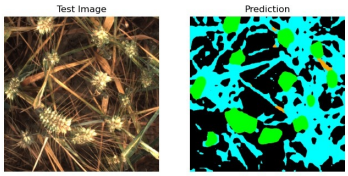
Conceptually, the methodology can be interpreted as a learning process in which the model repeatedly tries to generate segmentation masks and improves with each iteration. First, the model gives false predictions. But with comparisons to the correct masks and modifications to its internal parameters, it progressively learns to recognise patterns in various parts of plants.
Training an already trained model enables the network to begin with a good understanding of general visual attributes, which are then adapted to the specific task of wheat segmentation. The model learns to differentiate similar structures by sight by training through numerous training cycles and to label the correct structures at the pixel level.

III. Results and Discussion

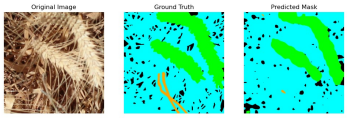
The trained model is shown to be capable of classifying wheat images into several semantic classes with fair precision. In practice, the model has learned meaningful representations of plant structures, as visual inspection indicates it has identified key plant structures, including spikes and leaves.



(a) Output from training set



(b) Output from test dataset



(c) Output from val dataset

Fig. 1: Example segmentation output showing successful identification of major plant structures such as spikes and leaves.

Here, as shown in the images, the stem classification was not very clear, as resizing removed the important information needed to identify it, which was getting cut off.

The performance of the proposed segmentation model was evaluated using standard metrics to assess its effectiveness in accurately identifying different wheat plant components.The

segmentation performance obtained from our experiments is as follows:

- **Training Performance:**
  - Pixel Accuracy: 0.9048
  - Class IoU: [0.8201, 0.8770, 0.4138, 0.8503]
  - Mean IoU (mIoU): 0.7403
  - Dice Score: 0.6752
- **Validation Performance:**
  - Pixel Accuracy: 0.8518
  - Class IoU: [0.7236, 0.7338, 0.0575, 0.7885]
  - Mean IoU (mIoU): 0.5758
  - Dice Score: 0.5109

The results show that the model achieves better performance on the training dataset compared to the validation dataset, indicating a generalization gap. While the model effectively learns to segment major structures such as spikes and leaves, the lower validation scores suggest challenges in handling unseen data, particularly for difficult classes like stems.

IV. Conclusion

For global wheat semantic segmentation we developed a deep learning-based framework, aiming to classify each pixel of an image into four categories: background, spike, stem, and leaf. The proposed approach utilizes a DeepLabV3 model with a ResNet-50 backbone, enabling effective extraction of both local and global features from agricultural images. The overall pipeline included data preprocessing, supervised training, prediction, and evaluation using standard segmentation metrics.

These results confirm that the proposed method is effective in capturing dominant structures such as spikes and leaves, while also improving the detection of previously challenging classes. However, the model still has certain limitations. It struggles to accurately classify thin structures such as stems, often failing to detect them completely. Additionally, the edges of leaves appear blurred, which reduces segmentation precision near boundaries.

In conclusion, the project successfully establishes a strong baseline for wheat semantic segmentation using deep learning. While the current model achieves promising results, further enhancements can significantly improve performance, especially for fine-grained structures, making it more suitable for real-world precision agriculture applications.

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